Maximum Average Subarrays

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. For each $i = 1, 2, \dots, n$, your task is to find the subarray ending at index i with the largest average. If there are multiple subarrays with the largest average, you should find the longest one.

Input

The first line has an integer n: the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print n integers: the length of the subarray ending at index i with the largest average for each $i = 1, 2, \dots, n$.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x_i \le 10^6$

Example

Input	Output
`7` `1 6 4 6 2 5 5`	`1 1 2 1 4 1 2`

Explanation: Consider $i = 5$. The averages of all subarrays ending at index 5 are $\frac{1+6+4+6+2}{5} = 3.8$, $\frac{6+4+6+2}{4} = 4.5$, $\frac{4+6+2}{3} = 4$, $\frac{6+2}{2} = 4$ and $\frac{2}{1} = 2$. The largest average is 4.5 and the length of the corresponding subarray is 4.